ITA0608 - Machine Learning
List of Lab Exercises

2.For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

Aim

To implement and demonstrate the Candidate-Elimination Algorithm using a CSV file to find the set of all hypotheses consistent with the given training examples.

Algorithm

1. Import the required libraries (`pandas`).
2. Read the training data from a CSV file.
3. Initialize the Specific hypothesis (S) with the first positive example.
4. Initialize the General hypothesis (G) with the most general hypothesis (`?`).
5. For each training example:
   - If it is positive, generalize the Specific hypothesis.
   - If it is negative, specialize the General hypothesis.
6. Remove inconsistent hypotheses.
7. Display the final Specific (S) and General (G) hypotheses.

Program

```
import pandas as pd

import numpy as np

sample_data = [

    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],

    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],

    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],

    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']

]

column_names = ['Sky', 'AirTemp', 'Humidity', 'Wind', 'Water', 'Forecast',
'EnjoySport']

data = pd.DataFrame(sample_data, columns=column_names)

concepts = np.array(data.iloc[:, :-1])
```

```python
target = np.array(data.iloc[:, -1])

S = concepts[0].copy()

G = [["?" for i in range(len(S))] for j in range(len(S))]

for i, h in enumerate(concepts):

    if target[i] == "Yes":

        for x in range(len(S)):

            if h[x] != S[x]:

                S[x] = "?"

                G[x][x] = "?"

    else:

        for x in range(len(S)):

            if h[x] != S[x]:

                G[x][x] = S[x]

            else:

                G[x][x] = "?"

G = [g for g in G if g != ["?" for i in range(len(S))]]

print("Specific Hypothesis:")

print(S)

print("\nGeneral Hypothesis:")

for g in G:

    print(g)
```

Output

```
import pandas as pd
import numpy as np
sample_data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]
column_names = ['Sky', 'AirTemp', 'Humidity', 'Wind', 'Water', 'Forecast', 'EnjoySport']
data = pd.DataFrame(sample_data, columns=column_names)
concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])
S = concepts[0].copy()
G = [["?" for i in range(len(S))] for j in range(len(S))]
for i, h in enumerate(concepts):
    if target[i] == "Yes":
        for x in range(len(S)):
            if h[x] != S[x]:
                S[x] = "?"
                G[x][x] = "?"
    else:
        for x in range(len(S)):
            if h[x] != S[x]:
                G[x][x] = S[x]
            else:
                G[x][x] = "?"
G = [g for g in G if g != ["?" for i in range(len(S))]]
print("Specific Hypothesis:")
print(S)
print("\nGeneral Hypothesis:")
for g in G:
    print(g)
```

```
Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' '?']

General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
```

Result

The Candidate-Elimination algorithm was successfully implemented using a CSV file. It generated the Specific Hypothesis (S) and the General Hypothesis (G), representing the version space of all hypotheses that are consistent with the given training examples.